

MATH 2400 - Week 1 Notes

1 Points vs. Vectors

We denote points by using round brackets: $P = (x_1, y_1)$. We think of points as representing **position**. The values in the brackets are called the **coordinates** of the point.

We denote vectors using angled brackets: $\vec{v} = \langle v_1, v_2 \rangle$. We think of vectors as representing **displacement**. The values in the brackets are called the **components** of the vector.

2 Dot Product

Given two vectors $\vec{v} = \langle v_1, v_2, \dots, v_n \rangle$ and $\vec{w} = \langle w_1, w_2, \dots, w_n \rangle$ the **dot product** (or inner product or scalar product) is defined to be

$$\vec{v} \cdot \vec{w} = \sum_{i=1}^n v_i w_i = v_1 w_1 + v_2 w_2 + \dots + v_n w_n$$

Please note that the dot product of two vectors is a scalar.

Algebraic Properties

1. $\vec{v} \cdot \vec{v} = |\vec{v}|^2$
2. $\vec{v} \cdot \vec{w} = \vec{w} \cdot \vec{v}$ (commutative)
3. $\vec{v} \cdot (\vec{w}_1 + \vec{w}_2) = \vec{v} \cdot \vec{w}_1 + \vec{v} \cdot \vec{w}_2$ (distributive)
4. $(c\vec{v}) \cdot \vec{w} = c(\vec{v} \cdot \vec{w}) = \vec{v} \cdot (c\vec{w})$ (associative under scalar multiplication)

Geometric Properties

The geometric interpretation of the dot product is

$$\vec{v} \cdot \vec{w} = |\vec{v}| |\vec{w}| \cos(\theta)$$

where θ is the angle between the vectors (in the plane determined by the vectors). Please note that $0 \leq \theta \leq \pi$.

This is often rewritten

$$\cos(\theta) = \frac{\vec{v} \cdot \vec{w}}{|\vec{v}| |\vec{w}|}, \quad \text{if } \vec{v}, \vec{w} \neq \vec{0}$$

This allows us to use the dot product to define that angle between two vectors.

By far the most important application of the dot product is using it to determine when two vectors are **orthogonal** (which is the grown-up name for perpendicular): two vectors are orthogonal if and only if $\vec{v} \cdot \vec{w} = 0$.

The second most important geometric application of the dot product is the **projection** of the vector $\vec{v}$ onto the vector $\vec{w}$, which is defined by

$$\text{proj}_{\vec{w}} \vec{v} = \left(\frac{\vec{v} \cdot \vec{w}}{|\vec{w}|} \right) \frac{\vec{w}}{|\vec{w}|} = \left(\frac{\vec{v} \cdot \vec{w}}{|\vec{w}|^2} \right) \vec{w} = \left(\frac{\vec{v} \cdot \vec{w}}{\vec{w} \cdot \vec{w}} \right) \vec{w}$$

Physical Properties

One of the very important physical interpretations of the dot product is that it allows us to find the **work** done by a force, $\vec{F}$ in displacing an object along a displacement vector, $\vec{D}$: $W = \vec{F} \cdot \vec{D}$.

3 Cross Product

Let $\vec{v} = \langle v_1, v_2, v_3 \rangle$ and $\vec{w} = \langle w_1, w_2, w_3 \rangle$ be two vectors in $\mathbb{R}^3$. Then the **cross product** (or outer product or vector product) is defined to be

$$\vec{v} \times \vec{w} = \langle v_2 w_3 - v_3 w_2, v_3 w_1 - v_1 w_3, v_1 w_2 - v_2 w_1 \rangle$$

This is impossible to memorize. Instead, use determinants:

$$\vec{v} \times \vec{w} = \det \begin{bmatrix} \vec{i} & \vec{j} & \vec{k} \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{bmatrix}$$

Algebraic Properties

1. $\vec{v} \times \vec{v} = \vec{0}$
2. $\vec{v} \times \vec{w} = -\vec{w} \times \vec{v}$ (anti-commutative)
3. $(c\vec{v}) \times \vec{w} = c(\vec{v} \times \vec{w}) = \vec{v} \times (c\vec{w})$ (associative under scalar multiplication)
4. $\vec{v} \times (\vec{w}_1 + \vec{w}_2) = \vec{v} \times \vec{w}_1 + \vec{v} \times \vec{w}_2$ (left-distributive)
5. $(\vec{v}_1 + \vec{v}_2) \times \vec{w} = \vec{v}_1 \times \vec{w} + \vec{v}_2 \times \vec{w}$ (right-distributive)

Geometric Properties

The most important geometric property of the cross product is that

$$(\vec{v} \times \vec{w}) \cdot \vec{v} = (\vec{v} \times \vec{w}) \cdot \vec{w} = 0$$

which says that the cross product of $\vec{v}$ and $\vec{w}$ is orthogonal to both $\vec{v}$ and $\vec{w}$. We use the **right-hand rule** to determine the direction of the cross product (relative to $\vec{v}$ and $\vec{w}$): with your right hand, point in the direction of $\vec{v}$ and curl your fingers towards $\vec{w}$. Then your right thumb points in the direction of $\vec{v} \times \vec{w}$.

Another important geometric property is that for any non-zero vector $\vec{v}$, $\vec{v} \times (k\vec{v}) = \vec{0}$. This says that the cross product between any two parallel vectors is the zero vector. It goes the other way as well: for any two non-zero vectors $\vec{v}$ and $\vec{w}$, if $\vec{v} \times \vec{w} = \vec{0}$, then $\vec{v}$ and $\vec{w}$ are parallel.

We also have that $|\vec{v} \times \vec{w}| = |\vec{v}||\vec{w}|\sin(\theta)$, where θ is the angle between $\vec{v}$ and $\vec{w}$. This says that the norm of $\vec{v} \times \vec{w}$ is equal to the area of the parallelogram spanned by the vectors.

Physical Properties

We can use the cross product to find **torque**. Imagine a wrench is turning a bolt (located at the origin). If the wrench is represented by a vector $\vec{r}$ and there is a force, $\vec{F}$, acting on the end of the wrench, then the torque, $\vec{\tau}$, on the bolt is given by $\vec{\tau} = \vec{r} \times \vec{F}$.